LANGMUIR FITTING: VISUAL & FLOWCHART GUIDE

Diagrams, flowcharts, and visual explanations for Phase 2

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1. OVERALL PHASE 2 WORKFLOW

```
                   PHASE 2: LANGMUIR FITTING
                    Input: 500 Samples Data

                             ↓

                   [1] LOAD DATA (500 × 13)

  Columns: Run, pH, C0, Time, Dose, Temp, Flow, Cl, Hard, CO3,
           NOM, Order, q_removal

  Data: 500 samples × 13 columns
  Response: q_removal (1.6 - 8.3 mg/g)

                             ↓

             [2] STANDARDIZE FEATURES & RESPONSE

  Input X: 500 × 10 (factors)
  Input y: 500 × 1 (q_removal)

  X_scaled = (X - mean) / std  →  mean=0, std=1
  y_scaled = (y - mean) / std  →  mean=0, std=1

  Output: X_scaled (500 × 10), y_scaled (500 × 1)

                             ↓

       [3] CREATE POLYNOMIAL FEATURES (Degree 2)

  Input:  10 factors (pH, C0, Time, ...)
  Add:    10 squared terms (pH², C0², Time², ...)
  Add:    45 interaction terms (pH×C0, pH×Time, ...)

  Output: 66 polynomial features
  Ratio:  500 samples / 66 features = 7.6 (good!)

                             ↓

       [4] FIT LINEAR REGRESSION MODEL
```

```
Minimize: Σ(y_scaled - ŷ)²

Method: Ordinary Least Squares (OLS)
 = (X^T X)^(-1) X^T y

Output: 67 coefficients (  + 66  's)
        (Linear regression model fitted!)

                           ↓

            [5] MAKE PREDICTIONS

ŷ_scaled = X_poly @    (matrix multiplication)

ŷ = scaler_y.inverse_transform(ŷ_scaled)
   (Convert back to original units: mg/g)

Output: 500 predictions (in mg/g)

                           ↓

          [6] CALCULATE PERFORMANCE METRICS

R² = 1 - (SS_res / SS_tot)           Expected: 0.84-0.87
RMSE = √(Σ(y - ŷ)² / n)             Expected: 0.9-1.2 mg/g
MAE = Σ|y - ŷ| / n                  Expected: 0.7-0.9 mg/g
Residuals = y - ŷ                   Expected: Normal,  0.92

Output: Metrics + Residuals (500 values)

                           ↓

          [7] CREATE DIAGNOSTIC PLOTS

Panel 1: Actual vs Predicted
Panel 2: Residuals vs Predicted
Panel 3: Residual Distribution
Panel 4: Q-Q Plot

Output: 4-panel PNG image

                           ↓

                  PHASE 2 COMPLETE

Output Files:
 langmuir_predictions.csv (500 × 15 columns)
 langmuir_model_info.json (metadata)
```

```
  langmuir_diagnostics.png (4 plots)

 Results:
  R²  0.85 (Langmuir baseline)
  RMSE  0.92 mg/g (prediction error)
  Residuals analyzed (ready for Phase 3)

 Next: Phase 3 - Residual Analysis & Feature Engineering
```

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2. DATA TRANSFORMATION PIPELINE

```
STAGE 1: INPUT DATA

 500 samples × 13 columns

Run | pH  | C0  | Time | ...  q_removal

1   | 4.4 | 2.7 | 87   | ...  4.23
2   | 4.1 | 3.0 | 97   | ...  3.87
... | ... | ... | ...  | ...  ...
500 | 7.5 | 7.0 | 33   | ...  3.49

               ↓
STAGE 2: EXTRACT FACTORS & RESPONSE

X: 500 × 10 (factors)
y: 500 × 1 (response)

               ↓
STAGE 3: STANDARDIZE (ZERO MEAN, UNIT VARIANCE)

X_scaled = (X - X.mean()) / X.std()
y_scaled = (y - y.mean()) / y.std()

Before: pH  [3, 9],  C0  [1, 10]
After:  pH  [-2, 2], C0  [-2, 2]
(All factors now have mean=0, std=1)

               ↓
STAGE 4: CREATE POLYNOMIAL FEATURES

Input: 10 factors
↓
Add main effects: 10 features
Add squared terms: 10 features (pH², C0², ...)
```

```
  Add interactions: 45 features (pH×C0, ...)
  ↓
  Output: 66 polynomial features

  X_poly shape: 500 × 66

                 ↓
STAGE 5: FIT REGRESSION MODEL

  LinearRegression().fit(X_poly, y_scaled)

  Learns: 67 coefficients (  + 66  's)

  Model: ŷ_scaled =   +  ×X_poly  + ...

                 ↓
STAGE 6: MAKE PREDICTIONS

  ŷ_scaled = model.predict(X_poly)
  ŷ = scaler_y.inverse_transform(ŷ_scaled)

  Output: 500 predictions in original units (mg/g)

                 ↓
STAGE 7: CALCULATE RESIDUALS

  residuals = y - ŷ

  Output: 500 residual values
  Expected: mean   0, std   0.92 mg/g
```

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3. THE LANGMUIR MODEL EXPLAINED

The Equation

```
                qmax × KL × Ce
            q =
                  1 + KL × Ce


Where:
  q    = Amount adsorbed at equilibrium (mg/g)
  qmax = Maximum adsorption capacity (mg/g)
  KL   = Langmuir binding constant (L/mg)
  Ce   = Equilibrium concentration (mg/L)
```

Shape of the Curve

```
Adsorption Amount (q)

                        qmax (saturation)

    qmax/2





                   KL = slope at origin
                        (higher KL = steeper)





        __________|____  Ce (Equilibrium concentration)

             1/KL
                        (Ce at qmax/2)
```

Key Points:
- At Ce = 0: q = 0 (no fluoride in solution = none adsorbs)
- At Ce = 1/KL: q = qmax/2 (half-saturation point)
- At Ce → ∞: q → qmax (approaches maximum)

Interpretation:
- qmax: How much carbon can hold (capacity)
- KL: How eager carbon is to adsorb (affinity)
- High KL: Steep curve (better binding)
- Low KL: Gradual curve (weaker binding)

How Factors Change the Curve

```
Base condition: pH=6.5, temp=25°C, no ions
                      q

          qmax=8.5



                            Ce

Effect of HIGH pH (pH=8):
                      q

          qmax=5.0
```

```
                   Ce
          (qmax decreased, KL decreased)

Effect of HIGH TEMPERATURE (temp=40°C):
                      q

          qmax=8.5




                   Ce
          (qmax same, KL increased, steeper curve)

Effect of HIGH IONS (Cl =100 mg/L):
                      q

          qmax=6.5


                   Ce
          (both qmax and KL decreased)
```

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4. FEATURE EXPANSION VISUALIZATION

From 10 Factors to 66 Features

```
ORIGINAL 10 FACTORS:

  1. pH
  2. C0 (initial concentration)
  3. Time
  4. Dose
  5. Temperature
  6. Flow Rate
  7. Chloride
  8. Hardness
  9. Carbonate
  10. NOM (Natural Organic Matter)

                    ↓
EXPAND TO 66 FEATURES:

GROUP 1: MAIN EFFECTS (10 features)
  pH
  C0
```

```
  Time
  Dose
  Temperature
  Flow Rate
  Chloride
  Hardness
  Carbonate
  NOM

GROUP 2: SQUARED TERMS (10 features)
  pH²           (captures bell curve: optimal at 6.5)
  CO²           (captures saturation effect)
  Time²         (captures approach to equilibrium)
  Dose²         (captures saturation)
  Temperature²  (captures curvature)
  Flow Rate²
  Chloride²
  Hardness²
  Carbonate²
  NOM²

GROUP 3: INTERACTION TERMS (45 features)
  pH × CO         (does concentration matter more at extreme pH?)
  pH × Time       (does time matter more at extreme pH?)
  pH × Dose       (does dose matter more at extreme pH?)
  ... [42 more]
  Carbonate × NOM (do ions and fouling interact?)

TOTAL: 1 (intercept) + 10 + 10 + 45 = 66 features
```

Why Each Type Matters

```
MAIN EFFECTS (pH, CO, Time, ...)

 Linear relationship with response

 Example:  q/ pH
 = "How much does q change per unit pH"

 Captures: Direct effect of each
 Factor magnitude and direction


SQUARED TERMS (pH², CO², Time², ...)

 Non-linear (curved) relationships

 Example: pH²
```

```
  = "pH effect is stronger far from 6.5"

  Captures: Bell curves, saturation,
  Inflection points, curvature


INTERACTION TERMS (pH×C0, pH×Time, ...)

  One factor modifies another's effect

  Example: pH × Time
  = "Effect of pH depends on time"

  Meaning: At short times, pH matters
  less. At long times, pH matters more.

  Captures: When one factor's effect
  depends on another factor's value
```

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5. MODEL FITTING VISUALIZATION

Before and After Scaling

```
BEFORE SCALING (Original Units):

  Factor values span different ranges:

  pH:    3 to 9        (range = 6)
  C0:    1 to 10       (range = 9)
  Time:  10 to 120     (range = 110)
  Temp:  20 to 40      (range = 20)
  NOM:   0 to 50       (range = 50)

  Problem:
  • Can't compare coefficients directly
  • Optimization is slow & unstable
  • Some coefficients are tiny (pH)
    others are large (Time)

                ↓
AFTER SCALING (Standardized):

  All factors have same range:

  pH_scaled:    -2 to 2   (std=1)
  C0_scaled:    -2 to 2   (std=1)
```

```
  Time_scaled:  -2 to 2   (std=1)
  Temp_scaled:  -2 to 2   (std=1)
  NOM_scaled:   -2 to 2   (std=1)

  Benefit:
  • Coefficients are comparable
  • Optimization is fast & stable
  • Can see relative importance


EXAMPLE COEFFICIENTS (Scaled):

   _pH = 0.15    (moderate effect)
   _CO = 0.42    (strong effect)
   _Time = 0.58  (strongest effect)
   _Temp = 0.08  (weak effect)

  Interpretation:
  Time is 5.8× more important
  than Temperature (per std dev)
```

The Fitting Process

```
INITIALIZATION:

  Random initial guess for  's
  Model prediction very poor
  Error (SSE) is huge

        ↓ [Fit using OLS]
ITERATION 1:

  Update  's to minimize error
  Error decreases
  (Closed-form solution, so done
   in 1 step! Not iterative.)

        ↓ [Final  's found]
CONVERGENCE:

  Final   coefficients found
  Error minimized
  R² calculated
  Done! (Happens in milliseconds)
```

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6. EXPECTED PERFORMANCE METRICS

R^2 Interpretation

```
R² = 0.85 (Expected)

    Explained Variance    85%



      Langmuir explains this much (good for Phase 2)

                        Residual Variance     15%

                          (ML will learn this in Phase 4)



Comparison to Other Models:

  Simple average (q_mean):      R²=0
  Simple 2-param Langmuir:      R² 0.50
  Our multi-factor Langmuir:    R² 0.85   (Expected)
  After ML correction:          R² 0.94 (Phase 5 target)
  Perfect fit:                  R²=1.00
```

RMSE Interpretation

```
RMSE = 0.92 mg/g (Expected)

Your data range: 1.64 - 8.32 mg/g

           8.5                       Maximum value

           7.0

           5.5           Data       Range = 6.68 mg/g
                        Range
           4.0
                         ----
           2.5

           1.0              Minimum value



RMSE = 0.92 represents average prediction error
```

```
If prediction is 4.2 mg/g:
  Actual could be: 4.2 ± 0.92 = [3.28 to 5.12] mg/g
  (68% confidence, ±1 )

Or: 4.2 ± 1.84 = [2.36 to 6.04] mg/g
    (95% confidence, ±2 )
```

```
Error as % of range:

  RMSE / Range = 0.92 / 6.68
               = 13.8%

  Interpretation:
    Excellent (< 10%): Ultra-precise
    Good (10-20%): Our range
  ~ Acceptable (20-30%): Workable
    Poor (> 30%): Needs improvement
```

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7. DIAGNOSTIC PLOTS INTERPRETATION

Panel 1: Actual vs Predicted

```
GOOD FIT:                                  BAD FIT:
Predicted                                  Predicted

 8                            8

 6                            6

 4            (follows line)  4          (above line)

 2                             2

 0            Actual      0           Actual
   0      2    4   6    8           0    2  4   6   8

Points close to line = Good predictions
Points far from line = Poor predictions
```

Panel 2: Residuals vs Predicted

```
GOOD (Random scatter):           BAD (Funnel shape):
Residual                         Residual

 2                                2
```

```
 0                       0

-2                              -2

 -4             Predicted -4            Predicted
     0    2   4   6   8                 0    2   4   6   8
```

```
GOOD:                           BAD:
• Random around zero            • Error increases with size
• Constant width                • Funnel pattern
• No patterns                   • Heteroscedasticity
• No curvature                  • May indicate model issue
```

Panel 3: Residual Distribution

```
GOOD (Normal distribution):     BAD (Skewed):
Frequency                       Frequency

 60                             60

 40                             40

 20                             20

  0                                    ---
   -2   -1   0   1   2                -2  -1  0  1   2
          Residuals                         Residuals
```

```
GOOD:                           BAD:
• Bell shape                    • Skewed right
• Centered at 0                 • Long tail
• Symmetric                     • May indicate outliers
• No extreme outliers           • Need investigation
```

Panel 4: Q-Q Plot

```
GOOD (Linear):                  BAD (S-shaped):
Sample                          Sample

 3                             3

 2                            2

 1                            1

 0                        0
```

```
-1                              -1

-2                              -2

-3            Theoretical-3            Theoretical

GOOD:                             BAD:
• Points follow line              • S-shape
• Minor tail deviations OK         • Non-normal
• Linear                          • Heavy tails
• Approximately normal            • May need transformation
```

---

8. EXPECTED OUTPUT FILES

File 1: langmuir_predictions.csv

```
Structure: 500 rows × 15 columns

Sample rows:
```

Run	pH	C0	Time	Dose	Temp	Flow	Cl	Hard	CO3	NOM	Order	q_rem	q_pred	residual
1	4.39	2.65	87	2.48	34.2	0.709	0	49	7	11	361	4.231	4.148	-0.083
2	4.12	2.99	97	4.1	35.7	0.651	63	280	1	15	73	3.872	3.923	+0.051
3	8.17	4.68	92	1.72	36.0	0.537	18	23	40	24	374	0.985	1.132	+0.147
..	...	...	...	...	...	...	...	...	...	...	...	...	...	...
500	7.51	7.02	33	2.75	28.5	1.725	61	376	55	16	102	3.487	3.542	+0.055

```
Use in Phase 3:
  • Identify patterns in residuals
  • Find where model struggles (largest |residual|)
  • Analyze by pH, time, dose, etc.
```

File 2: langmuir_model_info.json

```
{
  "phase": "Phase 2: Langmuir Fitting",
  "date": "2026-05-03",
  "n_samples": 500,
  "n_factors": 10,
  "n_features_original": 10,
  "n_features_expanded": 66,
  "model_type": "Multi-factor Langmuir (Polynomial degree 2)",
  "scaling": "StandardScaler (both X and y)",
  "performance_metrics": {
    "R2": 0.8456,           ← Most important
```

```
    "RMSE": 0.9231,         ← Prediction error
    "MAE": 0.7145,          ← Mean absolute error
    "residual_mean": -0.000001,  ← Should be ~0
    "residual_std": 0.9228  ← Residual spread
  }
}
```

Use in Phase 3-5:

- Reference baseline metrics
- Compare with Phase 4 ML model
- Track improvement to hybrid model

File 3: langmuir_diagnostics.png

4-panel diagnostic plot showing:

```
Panel 1: Actual vs Predicted       Panel 2: Residuals vs Predicted
  (Should follow diagonal)           (Should scatter around 0)
      8                                  2

                                     0

    0                                  -2


Panel 3: Residual Distribution    Panel 4: Q-Q Plot
  (Should be bell curve)           (Should be linear)






       -2  0   2
```

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9. SUMMARY TABLE

```
                       PHASE 2 SUMMARY

 ASPECT                    DETAILS

 INPUT                     500 samples, 10 factors, q_removal
 PROCESS                   Standardize → Polynomial → Fit
 MODEL                     Multi-factor Langmuir (66 features)
 METHOD                    Linear Regression (OLS)
 PARAMETERS                67 coefficients (  +  ...  )
 EXPECTED R²               0.84-0.87 (84-87% explained)
```

```
EXPECTED RMSE            0.9-1.2 mg/g
EXPECTED MAE             0.7-0.9 mg/g
RESIDUALS                Normal, random,  0.92 mg/g
OUTPUT FILES             CSV, JSON, PNG
RUNTIME                  < 1 minute (mostly for visualization)
STATUS                   Baseline model (R²  0.85)
NEXT PHASE               Phase 3: Residual Analysis
```

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10. PHASE 2 → PHASE 3 TRANSITION

After Phase 2:

```
                    BASELINE ESTABLISHED

   Langmuir model fitted
   R² = 0.85 (What chemistry can do)
   Residuals = 0.92 mg/g (What ML needs to learn)
   500 predictions made
   Diagnostic plots created

 Question for Phase 3:
 "Where are the patterns in residuals?"
 "What is the model missing?"
 "What can ML learn?"

                         ↓
                   PHASE 3 WORK:
       Analyze residuals by factor combinations
       Find non-linear patterns chemistry missed
       Engineer features to capture those patterns
       Prepare training data for ML models
                         ↓
                  PHASE 4 WORK:
       Train Random Forest, XGBoost, MLP on residuals
       Cross-validate (5-fold CV)
       Select best model
       Expected: R²  0.90-0.93 on residual prediction
                         ↓
                  PHASE 5 WORK:
       Hybrid = Langmuir + ML residual prediction
       Combine: q_hybrid = q_langmuir + q_ml_correction
       Achieve: R²  0.94 (Target!)
                         ↓
        SUCCESS: Hybrid model 20-35% better than chemistry!
```

End of Visual & Flowchart Guide

This document provides visual representations, flowcharts, and diagrams for understanding Phase 2 Langmuir fitting.

EOF cat /mnt/user-data/outputs/LANGMUIR_VISUAL_GUIDE.md